STA 445 S26 Assignment 5

2026-02-26

Problem 1

Convert the following to date or date/time objects.

a. September 13, 2010.

```
mdy("September 13, 2010")
```

```
## [1] "2010-09-13"
```

b. Sept 13, 2010.

```
make_date(year=2010, month=9, day=13)
```

```
## [1] "2010-09-13"
```

c. Sep 13, 2010.

```
mdy("Sep 13, 2010")
```

```
## [1] "2010-09-13"
```

d. S 13, 2010. Comment on the month abbreviation needs.

```
make_date(year=2010, month=9, day=13)
```

```
## [1] "2010-09-13"
```

If not a number, the month needs to be the full name or the standard 3 letter abbreviation, ex not S or Sept.

e. 07-Dec-1941.

```r
dmy("07-Dec-1941")
```

```
## [1] "1941-12-07"
```

f. 1-5-1998. Comment on why you might be wrong.

```r
dmy("1-5-1998")
```

```
## [1] "1998-05-01"
```

```r
mdy("1-5-1998")
```

```
## [1] "1998-01-05"
```

5 and 1 could both be month or day, so there's two possibilities.

g. 21-5-1998. Comment on why you know you are correct.

```r
dmy("21-5-1998")
```

```
## [1] "1998-05-21"
```

While 5 is ambiguous, 21 must be day, so 5 must be month.

h. 2020-May-5 10:30 am

```r
ymd_hm("2020-May-5 10:30 am")
```

```
## [1] "2020-05-05 10:30:00 UTC"
```

i. 2020-May-5 10:30 am PDT (ex Seattle)

```r
ymd_hm("2020-May-5 10:30 am PDT", tz="America/Los_Angeles")
```

```
## [1] "2020-05-05 10:30:00 PDT"
```

j. 2020-May-5 10:30 am AST (ex Puerto Rico)

```r
ymd_hm("2020-May-5 10:30 am AST", tz="America/Puerto_Rico")
```

```
## [1] "2020-05-05 10:30:00 AST"
```

Problem 2

Using just your date of birth (ex Sep 7, 1998) and today's date calculate the following:

a. Calculate the date of your 64th birthday.

```r
bday <- mdy("Jan 28 2006")
bday + years(64)
```

```
## [1] "2070-01-28"
```

b. Calculate your current age (in years).

```r
age <- time_length(interval(bday, today()), "years")
```

c. Using your result in part (b), calculate the date of your next birthday.

```r
next.bday <- bday + years(floor(age + 1))
```

d. The number of *days* until your next birthday.

```r
next.bday - today()
```

```
## Time difference of 336 days
```

e. The number of *months* and *days* until your next birthday.

```r
as.period(interval(today(), next.bday), unit="month")
```

```
## [1] "11m 2d 0H 0M 0S"
```

Problem 3

Suppose you have arranged for a phone call to be at 3 pm on May 8, 2015 at Arizona time. However, the recipient will be in Auckland, NZ. What time will it be there?

```r
call.date <- ymd_hm("2015 May 8 3:00 PM", tz="US/Arizona")
with_tz(call.date, "Pacific/Auckland")
```

```
## [1] "2015-05-09 10:00:00 NZST"
```

Problem 4

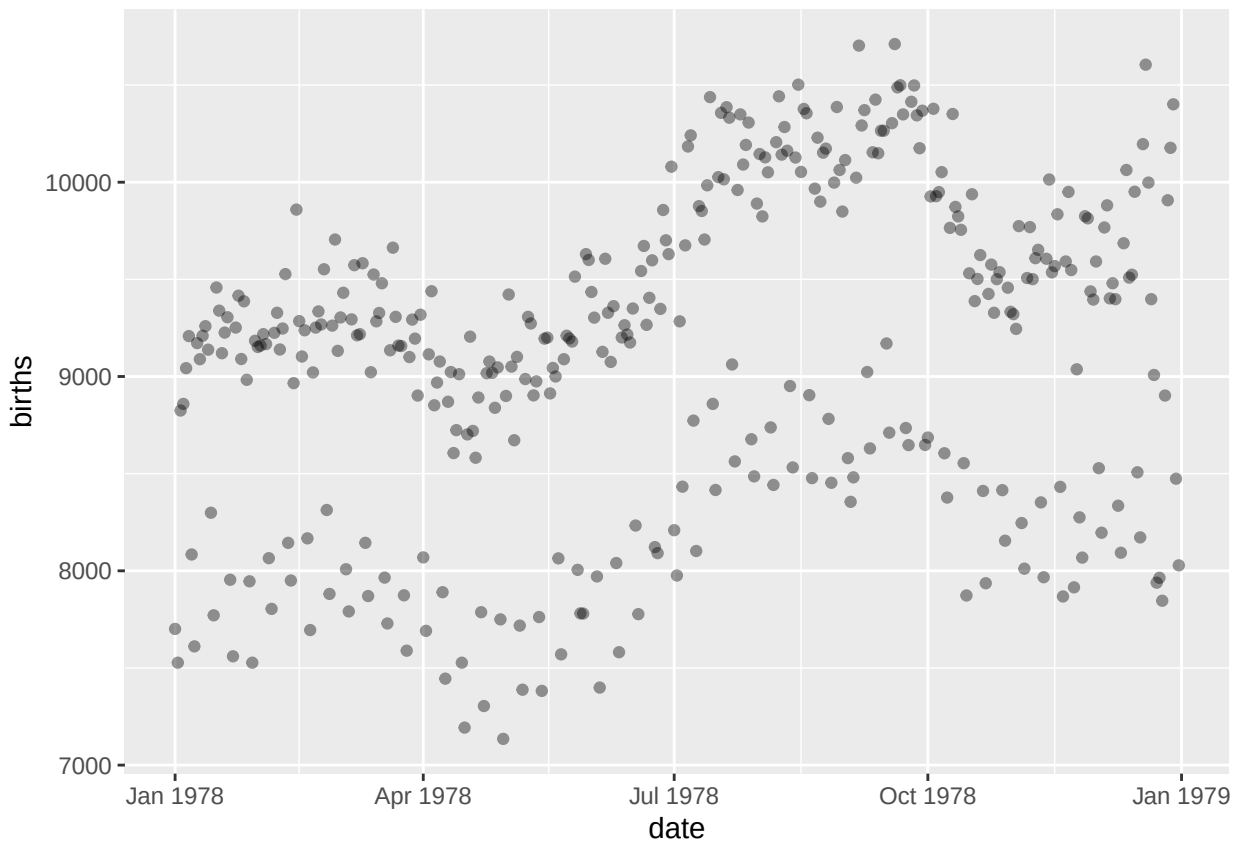
It turns out there is some interesting periodicity regarding the number of births on particular days of the year.

a. Using the `mosaicData` package, load the data set `Births78` which records the number of children born on each day in the United States in 1978. Because this problem is intended to show how to calculate the information using the `date`, remove all the columns *except* `date` and `births`.

```
Births <- Births78 %>% select(contains("date") | contains("birth"))
```

b. Graph the number of `births` vs the `date` with date on the x-axis. What stands out to you? Why do you think we have this trend?

```
ggplot(Births, aes(x=date,y=births)) + geom_point(alpha=.4)
```



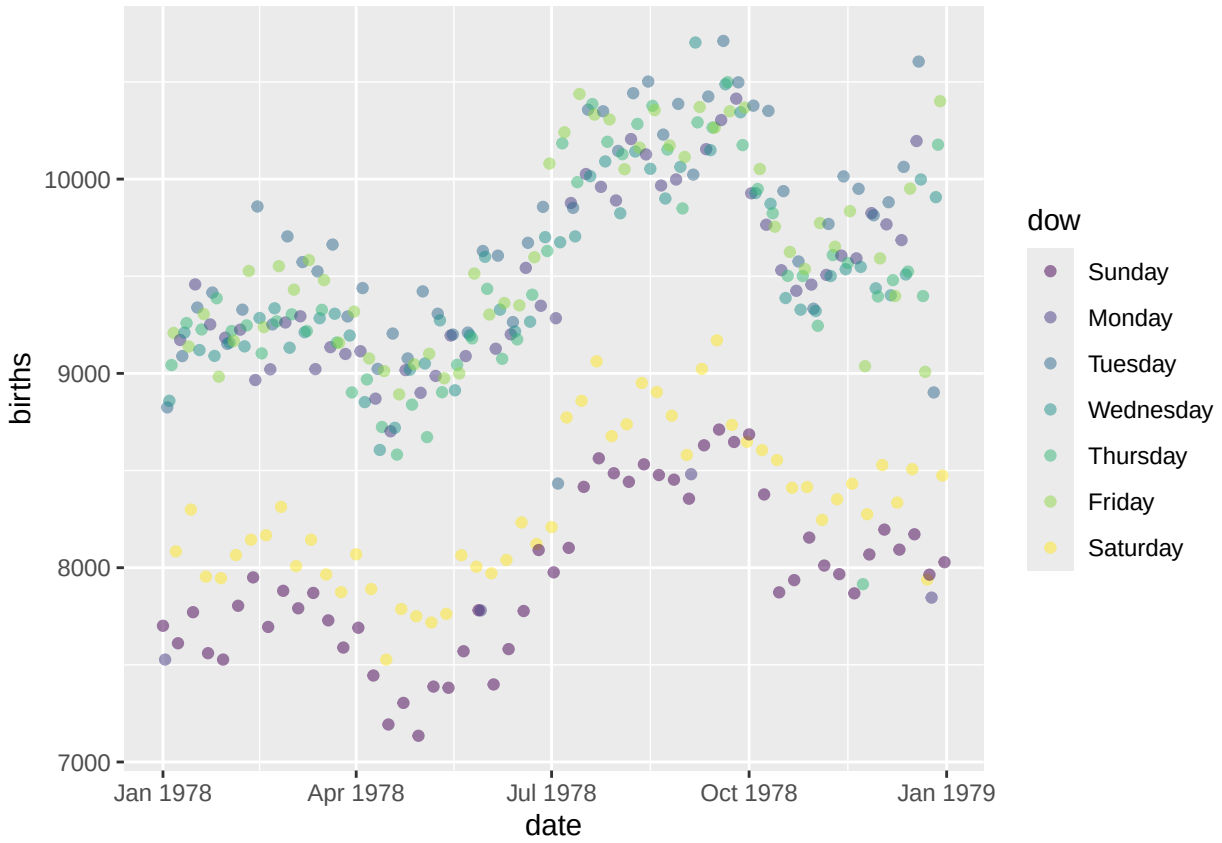
Some points are about 1000 below points around the same date, forming two distinct bands. Weekends might be the cause, as doctors probably don't work weekends as much and might induce or delay in order to avoid weekends.

c. To test your assumption, we need to figure out the what day of the week each observation is. Use `dplyr::mutate` to add a new column named `dow` that is the day of the week (Monday, Tuesday, etc). This calculation will involve some function in the `lubridate` package and the `date` column.

```
Births <- Births %>% mutate(dow=wday(date, label=TRUE, abbr=FALSE))
```

d. Plot the data with the point color being determined by the day of the week variable.

```
ggplot(Births, aes(x=date,y=births, colour = dow)) + geom_point(alpha=.5)
```


10000
9000
8000
7000
births
Jan 1978
Apr 1978
Jul 1978
Oct 1978
Jan 1979
date
dow
Sunday
Monday
Tuesday
Wednesday
Thursday
Friday
Saturday